MILITARY SPECIFICATION

ELECTRON TUBE: 7688738
(INFRARED, IMAGE CONVERTER)

1. SCOPE

1.1 This specification covers one type of image converter, infrared-sensitive, electron tube. The image tuve is equipped with an infrared sensitive S1 Photo-cathode, P-20, phosphor image screen, a potted optical lens, and three electrical connecting leads.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-1	Electron Tubes: General-Specification for
MIL-0-13830	Optical Components for Fire-Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Munitions Command

C7688738	Electron Tube (Infrared Image Converter)
F8584978	Universal Standard Image Tube Test Fixture
IEL 7688738	List of inspection Documents for Electron Tube: 7688738

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PACKAGING DATA SHEET

MIL-P-14232 P7688738 Packaging of Electron Tube: 7688738

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Initial production sample. - Requirements for the submission of an initial production sample by the contractor shall be as specified in the contract.

3.2 Materials. - Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2.1 Glass coating. - Two coats of a solution of 40 cubic centimeters of Dow-Corni,ng Silicone Resin 2920 (or equal) in 3600 cubic centimeters of toluene shall be applied to the exposed glass surface of the image tube with the exception of the photocathode and image screen face plates.

3.2.2 Fluorescet material. - The image screen shall be aluminized and coated with Type P-20 Phosphor (see 6.4.5).

3.3 Fabrication. - The image tube shall be manufactured in accordance with Drawing C7688738 and drawings pertaining thereto.

3.3.1 Terminals. - The anode, cathode, and focusing electrodes at the soldered joints, along with all other processing electrodes, shall be coated with glyptol (General Electric Glyptol No. 1201, or equal) to a mininum 1/16 inch radius around each terminal.

3.3.2 Potting. - The potting compound shall not contain voids or bubbles so aligned as to reduce the dielectric strength or adherence to the glass envelope.

3.3.3 Pot surfaces. - All pot surfaces shall be free of cracks and flash. Scuffs on the external pot surface shall not exceed an area equivalent to that of a 3/32 inch square.

3.3.4 Manufacturer’s stabilization period. - The tube shall be held for a manufacturer’s stabilization period of at least 336 hours before meeting the requirements of the specification.

3.4 General specifications. - The contractor shall be responsible for adherence to and compliance with the requirements specified below under each respective general specification.

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3.4.1 MIL-E-1.

a. Abbreviations and symbols
b. Material
c. Absolute maximum ratings

3.4.2 MIL-O-13830.

a. Materials
b. Cleanliness
c. Lenses

3.4.3 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship
f. Definitions

3.5 Environmental requirements.

3.5.1 Temperature cycling and thermal shock.- The image tube shall meet the remaining requirements of this specification, and shall show no evidence of deformation, cracks, or breakage, after exposure to the following ambient temperature excursions in uninterrupted sequence:

a. A gradual decrease from 70° ± 15°F to -80°F; maintained for at least one hour
b. A gradual increase to -25°F; maintained for at least one hour
c. Immediate exposure to 75° ± 15°F; maintained for at least one hour
d. A gradual increase from 75° ± 15°F to 160°F; maintained for at least one hour
e. A decrease to 125°F; maintained for at least one hour
f. Immediate exposure to 75° ± 15°F

3.5.2 Vibration. - Prior to vibration, the image tube shall meet the requirements of 3.8.4 and 3.8.5. The image tube shall then show no evidence of physical damage and shall meet the following requirements of this specification, after being subjected to the requirements of 3.5.2.1 and 3.5.2.2 with the following prevalent conditions:

a. At an ambient temperature of 75° ± 15°F
b. With no electrical potential applied across the image tube
c. When rigidly vertically mounted with the photocathode end up
d. When vibrated at each specified frequency for at least 2 minutes

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3.5.2.1 Vibration: Thirty cycle.- A simple harmonic motion having an amplitude of 0.0625 ± 0.0025 inch (total excursion of 0.125 ± 0.005 inch) at a frequency of 30 ± 2 cycles per second (cps).

3.5.2.2 Vibration: Fifty cycle.- A simple harmonic motion having an amplitude of 0.040 ± 0.0025 inch (total excursion of 0.080 ± 0.005 inch) at a frequency of 50 ± 2 cps.

3.5.3 Shock.- Subsequent to meeting requirement 3.5.2 the image tube, in proper holder, shall not exhibit flashing or electrical bursting during more than two shocks in each specified direction. Six shocks shall be applied parallel with and six perpendicular to its longitudinal axis (total of 12 shocks). Each shock shall be applied at 75° ± 15°F and shall have the following characteristics:

a. A half-sine wave pulse of at least 75 gravitional units (G's) at its peak
b. A pulse duration of 6 ± 2 milliseconds between the 10% values of its peak amplitude
c. The after oscillations shall not exceed 15% of the nominal half-sine wave peak amplitude

3.6 Pot and lens cleanliness and lens surface quality.- When the image tube focusing voltage (Ebl) is adjusted for optimum resolution (during which the entire photocathode is illuminated for best contrast), examination of the areas specified in 3.6.1 through 3.6.3 (by 10 power magnification) shall be in accordance with the criteria defined in MIL-O-13830. All circles shall be concentric to the tube's geometric axis (see 6.4.6), as viewed through the field lens and subject to the magnification thereof.

3.6.1 Field lens quality.- The field lens shll have a minimum surface quality of 40-20 within an 0.80 inch diameter circle.

3.6.2 Screen lens quality. - Within a 0.25 inch circle (inner zone) the image screen shall have at least a 30-10 surface quality. The intermediate zone-on the image screen between a 0.25 and 0.65 inch circle shall have a minimum surface quality of 40-20. The inner and intermediate zones shall contain no more than one particle of dirt of a size greater than a Number 10 dig nor more than one piece of lint in excess of one mil (1/1000 inch) in width and 30 mils in length. If more than one piece of lint or scratch exists in the inner zone, the two together shall not exceed one and one-half times the size specified for one.

3.6.3 Pot cleanliness. - Screen area beyond the 0.65 inch circle and pot cavity shall not contain loose dirt or lint particles, exceeding the number and size previously specified in 3.6.2, that may under shock, vibration, or static charge, relocate in the inner or intermediate zones. Potting resin shall not extend into the 0.65 inch diameter area. There shall be no evidence of chip-out at the junction of the field lens ground edge and potting resin. Seals shall exhibit no more than 0.0060 inch of separation. There shall be no evidence of lens glass fracture nor loose gaskets between the pot and field lens, or evidence of dust retention grease.

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3.7 Electrical conditions. - Unless otherwise specified in the Individual operational requirements (3.8), the conditions defined in 3.7.1 through 3.7.7 shall prevail.

3.7.1 Photocathode polarity. - The photocathode polarity of the image tube respective to the image screen shall be negative.

3.7.2 Photocatode potential. - The direct current voltage (Vdc) potential (Eb2) at the image tube photocathode shall be 16,000 + 100 volts.

3.7.3 Stabilization time. - The image tube shall be permitted a stabilization time of 15 seconds after the application of its operating potential.

3.7.4 Light source. - Unless otherwise specified, the radiation source shall be obtained from a tungsten filament lamp operating at a color temperature of 2870° Kelvin (K).

3.7.5 Filter. - The light source specified in 3.7.4 shall be filtered by Corning No. 2540 Infrared Filter (Melt No. 1613) of 2.61 millimeter thickness or equivalent.

3.7.6 Quantity of radiation. - The quantity of radiation from the source incident on the photocathode for the individual tests shall be the amount stated in the individual operational requirement.

3.7.7 Absolute maximum rating. - The absolute maximum ratings of the image tube shall be as specified in Table I.

TABLE I - ABSOLUTE MAXIMUM RATINGS

Parameter:	Eb2	Eb1	Light Flux	ib (see 3.7.7.1)	Ib	TA	ALT
Unit:	Vdc	Vdc	Lumen	µAdc	µAdc	°F	ft
Maximum:	17,000	3,375	----	3.5	0.35	+160	10,000
Minimum:	----	----	----	---	----	-80	----

3.7.7.1 Maximum current (ib). - The period of time of ib, in dc microampere (uAdc), for any single period shall not exceed 2 minutes.

3.7.8 Focus. A 25 line-pair-per-millimeter pattern at the cathode shall focus on the center of the image screen with an applied dc potential (Eb1) of between 2,120 and 2,600 volts.

3.8 Operational requirements. - Unless otherwise specified, the operational requirements shall be met at an ambient temperature of between 60° to 90°F and after the holding period specified in 3.8.2.

3.8.1 Conversion index. - The image tube shall exhibit a minimum conversion index of 15, when the radiation source specified in 3.7.4 is used to produce a 0.75 ± 0.05 inch diameter circular image on the center of the photocathode and the amount of the incident flux on the photocathode (prior to interposing a filter) is 0.1 ± .005 lumen. The conversion index or measurement of the overall sensitivity of the image tube shall be defined as:

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$$CI = \frac{Fe\ K}{Fit}$$

Where: a. CI = Conversion index
b. K = Correction factor to compensate for the light loss between screen and lucite cone resulting from interposition of the field lens (K = 2.85)
c. Fe = The total available luminous flux emitted by the image screen at the operating voltage specified herein as collected by the lucite cone.
d. Fi = The unfiltered 2870°F flux incident on the photocathode of the image tube.
e. T = The filter factor (see 6.4.1) of the interposed infrared filter (see 3.7.5).

Conversion index shall be met before the holding period specified in 3.8.2.

3.8.2 Conversion index decay. - After a holding period of 672 hours, which shall not include the manufacturer's stabilizing period (at least 336 hours), the conversion index of the tube shall not be less than 15 and shall not have slumped more than 20 percent from the initial value measured at the end of the stabilizing period. If the conversion index has slumped more than 20 percent, but still is above the minimum value of 15, the tube shall be held an additional 336 hours. At the end of this 1008 - hour period, the tube shall have a conversion index of not less than 15 and shall not have slumped more than 30 percent of the initial value measured at the end of the stabilization period. The date of conversion index measurement, the letters CI, and the value of the conversion index obtained shall be marked on each tube (see 6.5).

3.8.3 Conversion index to background.- The conversion index to background shall not be less than 4, when defined as the ratio: $\frac{CI}{LB}$ when measured at an ambient temperature of 75 + 5°F.

Where: CI = Conversion index
LB = Background brightness (micro-lamberts).

3.8.4 Alinement. - The center of a reference located on the image tube photocathode at its geometric axis (see 6.4.6), shall be imaged within 0.0375 inch of the geometric axis of the image tube screen field lens.

3.8.5 Image shift. - Subsequent to meeting the requirements of 3.8,4, the position of the imaged point, initially recorded therein, shall not be displaced more than 0.0020 inch in any direction, due to the electrical shock specified herein. There shall be no evidence of flicker or blooming nor shift or drift from the initial position when observed for at least one minute. The electrical shock shall consist of 10 cycles of alternate periods of, 15 seconds with and 10 seconds without application of the image tube rated voltage (see 3.7.2, 3.7.8). This requirement shall be met prior to and after meeting requirements of 3.5.2 and 3.5.3.

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3.8.6 Resolution. - The image tube shall meet the following requirements, when the input radiation level is adjusted for maximum contrast and the focusing voltage (Ebl) is between 2120 and 2600 Vdc.

3.8.6.1 Center resolution. - The image screen shall resolve a 25 line-pairs-per-millimeter (l-p/mm) pattern of the photocathode in all four meridians within a 0.075 inch radius circle concentric with the image tube geometric axis.

3.8.6.2 Peripheral resolution. - The image screen shall resolve each quadrant of an 8 l-p/mm pattern on the photocathode in all four meridians within the area of a 0.04 inch square. The center of this square shall be 0.3 inch from the center of the geometric axis of the tube.

3.8.7 Cathode and screen spots. - There shall be no ion spots or cold emission, appearing as bright streaks or blemishes, when the tube is viewed with 2.5 power magnification and with no radiation incident to the photocatode. Opaque or dark spots shall not exceed the size and quantities, specified in Table II, when the screen is viewed by 10 power magnification and the level of radiation on the photocathode and the focusing voltage are adjusted for optimum spot contrast.

TABLE II

Size of spots 1/ (in millimeters)	Number of spots within 0.25 inch diameter circle 2/	Number of spots, within area bounded by two circles 0.25 and 0.57 inches in diameter 2/
Greater than 0.40	0	0
0.30 to 0.40	0	2
0.20 to 0.30	1	10
0.08 to 0.20	7	20
Max. Total	7	28

1/ If the distance between two spots is less than the maximum dimension of either spot they shall be considered as one spot with a size equal to the sum of the maximum dimensions of the two spots plus the amount of separation between them.

2/ The 0.25 and 0.57 inch circles on the image screen shall be concentric with the tube axis (see 6.4.6).

3.8.8 Center magnification.- Center magnification (Cmx) shall be 0.580 ± .010 for 2 object points on the cathode, equidistant to the geometric axis, 0.0805 ± 0.0001 inch apart when meeting 3.7.8.

3.8.9 Percentage distortion.- Percentage distortion shall be between 8 and 12 percent as determined by the following formula:

$$D = \left(\frac{M_2 - M_1}{M_1} \right) \times 100$$

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Where:

D = Percentage distortion
M1 = Center magnification
M2 = Edge magnification measured with 2 object points, equidistant to the geometric axis, 0.3675 ± 0.0001 inch apart

3.8.10 Cathode and screen uniformity.- The background shading shall be uniformly graded so that there is no line of demarcation between light and dark areas under the following conditions:

a. With no radiation incident on the photocathode
b. With 0.1 to 0.4 lumens/sq. ft. of radiation distributed uniformly over the photocathode. The radiation specified in 3.7.4 need not be used for this requirement.

3.8.11 Breakdown. - The image tube shall show no evidence of intermittent or continuous arcing, spraying, or flashing when the focusing voltage is slowly swept from 2120 to 3375 Vdc, with an applied potential of 17,000 $^{+0}_{-100}$ Vdc for no less than one minute. The image tube shall meet this requirement with a room illumination of not more than 0.01 foot candle.

3.8.12 Dark current. - The dark current drawn by the image tube shall not exceed 1 microampere (ua) with zero light flux (LIK) on the photocathode. This requirement shall be met under a condition of 40% to 50% relative humidity (RH) at an ambient temperature of 75 ± 5°F.

3.8.13 Irradiation surge.- The screen image shall not disintegrate, for any part darken or blank-out, when the cathode is subjected to an irradiation surge from a tungsten light source, not necessarily the one specified in 3.7.4, to produce at least 3.5 uAdc of cathode current for at least 0.5 second.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 Initial production test. - Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.1 Initial production sample failure.- Should one unit fail to meet the initial production testing specified in 4.2, the contractor shall cease those operations which are causing the rejection and make the necessary corrections thereto. When corrections in the process or controls have been made, two additional (new) production models shall be submitted in accordance with 4.2.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless othewise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with “Submission of Product” provisions of MIL-STD-l05.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Tables III, IV and V herein. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE III - CLASSIFICATION OF DEFECTS

CRITICAL: NONE DEFINED MAJOR: AQL 0.40% Defective		REQUIREMENTS	TEST PROCEDURES
101.	Alinement, pre-vibration - shock	3.5.2	4.6.6
102.	Image shift, pre-vibration - shock	3.5.2	4.6.6
103.	Physical failure, post vibration	3.5.2	4.6.7.1
104.	Electron function, shock testing	3.5.3	4.6.8
105.	Conversion index	3.8.1	4.6.10
106.	Conversion index decay	3.8.2	4.6.11
107.	Conversion index to background	3.8.3	4.6.12
108.	Alinement, post vibration - shock	3.8.4	4.6.13
109.	Image shift, post vibration - shock	3.8.5	4.6.14
110.	Resolution	3.8.6	4.6.15
111.	Magnification and focus	3.8.8	4.6.17
112.	Percentage distortion	3.8.9	4.6.18
113.	Breakdown	3.8.11	4.6.20
114.	Dark current	3.8.12	4.6.21
115.	Irradiation surge	3.8.13	4.6.22

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TABLE III - CLASSIFICATION OF DEFECTS (Continued)

MAJOR: AQL 0.65% Defective	REQUIREMENTS	TEST PROCEDURES
116. Cleanliness; lens, screen, pot	3.6	4.6.9
117. Surface quality; lens, screen, and potting	3.6	4.6.9
118. Cathode and screen spots	3.8.7	4.6.16
119. Cathode and screen uniformity	3.8.10	4.6.19

MINOR: NONE DEFINED

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General. - One image tube shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table IV.

TABLE IV

CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURES
301. Fabrication	3.3	4.6.4 - per drawing
302. General specifications	3.4	MIL-E-1, MIL-O-13830, MIL-F-13926
303. Thermal shock	3.5.1	4.6.5

4.4.2 Environmental. - Three image tubes shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests in Table 111 and shall then meet the requirements and tests in Table V.

TABLE V

CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURES
304. Glass coating	3.2.1	4.6.2.1
305. Fluorescent material	3.2.2	4.6.3.1

4.4.3 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government furnished inspeciton equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in IEL 7688738.

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the “Test Methods and Procedures” specified in 4.6 of this specification and with the component inspection procedures specified in “Examination” and “Test Facilities” requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concetp, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the Prescribed tolerance limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Auxiliary inspection equipment. - Fixture 7560125 is classified as an auxiliary inspection fixture to Fixture 8584978, and is used to expedite acceptance inspection for the specified characteristics. Acceptance inspection on the auxiliary fixture is predicated on its calibration to Fixture 8584978 for the applicable characteristic. Calibration shall be conducted daily or prior to use, as applicable. The recorded values of an acceptable image tube inspected on Fixture 8484978 shall be utilized for calibration of Fixture 7560125.

4.6 Test methods and procedures.

4.6.1 Manufacturer’s stabilization time.- The image tube shall conform to the requirements of 3.3.4 prior to acceptance inspection for the following characteristics.

4.6.2 Glass coating. - The completeness of the coating material on the glass surface of the image tube shall be visually inspected, with the aid of at least 10 power magnification, for conformance to the requirements of 3.2.1. The material used, and the number of coats applied to the specified glass surfaces of the image tube shall be certified for each lot submitted for Government acceptance.

4.6.3 Fluorescent material.- When the image tube is energized as in 4.6.3 the phosphor coating on the image screen shall luminesce in the yellow-green area of the visible spectrum. The material used for coating the image screen of the tube shall be certified for each lot submitted for Government acceptance.

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4.6.4 Fabrication. - The image tube shall be subjected to a dimensional and visual inspection. The image tube shall conform to the requirements of 3.3, 3.3.1, 3.3.2 and 3.3.3. Standard measuring equipment shall be utilized for dimensional inspection. The material used for coating the terminals of the image tube shall be certified for each lot submitted for Government acceptance.

4.6.5 Thermal shock. - Test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.5.1. Upon return to standard ambient temperature the image tube shall be subjected to a visual and tactile examination and shall show no evidence of deformation, cracks, or breakage. The imgage tube shall then be subjected to the following tests, and shall conform to the requirements specified.

4.6.6 Alinement and image shift, previbration. - These tests shall be conducted subsequent to the holding period specified in 3.8.2. The image tube shall conform to the requirements of 3.8.4 and 3.8.5, prior to the vibration and physical shock tests, when tested in accordance with 4.6.13 and 4.6.14 respectively.

4.6.7 Vibration. - Standard vibrating equipment capable of providing the amplitude and frequency specified for low frequency vibration(3.5.2.1) and that specified for high frequency vibration in 3.5.2.2 shall be utilized for this test. The image tube shall be vibrated to the conditions specified in 3.5.2. The vibrating equipment shall be positioned so as not to transmit any vibration to the electronic test equipment used for testing the performance requirements of the image tube.

4.6.7.1 Post vibration. - Subsequent to vibration the image tube shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.5.2 prior to being subjected to the remaining tests specified herein. A visual examination shall be made for all missing or damaged parts including damage to the pot assembly, image screen, photocathode, wires and terminals, prior to performing the performance tests herein. A tactile examination should also be conducted to determine looseness of the above accessible items. Optical characteristics shall be examined by observing through the potted lens to detect excessive dirt, obstruction or interference with theoptical system that would be cause for rejection. An audible examination shall be made by positioning the image tube close to the ear and vigorously shaking it with the hand while listening for loose internal optical, mechanical and electronic parts (lens, image screen, terminals, photocathode, etc.). Any one of the possible defects defined above shall be cause to consider an image tube defective unless separate defects - per-hundred type sampling plans are established and approved by the Government as part of the supplier's inspection system. Image tubes which have passed the above examinations shall be subjected to the tests of 4.6.8 to 4.6.21 inclusive to determine overall compliance with 3.5.2.

4.6.8 Shock. - This test shall be performed subsequent to the vibration test. Shock testing equipment that conforms to Drawing D7680617 with an adapter that conforms to Drawing F7681036 shall be utilized for this

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test. The voltage to energize the image tube shall be supplied by a power source that is capable of maintaining a constant voltage to the appropriate terminal, within the limits of 3.7.2 and 3.8.6. There shall be no incident light on the photocathode of the image tube during this test. The visual inspection for flashing and electrical bursting in the image tube, during shock application, shall be conducted with the level of daylight interference of observation reduced to the lowest possible level. Position and energize the image tube in the shock test fixture, adjust the focus input voltage to the tube to the mean voltage value, and permit the tube a 15 second stabilization period before shock testing. Subject the energized image tube to the shock impacts specified in 3.5.3. The presence of flashing or electrical bursting during more than two shocks in each direction shall be cause for rejection. The shock testing equipment shall be positioned so as not to transmit any shock or vibration to the electronic test equipment used for testing the performance requirements of the image tube.

4.6.9 Cleanliness and surface quality.- This test shall be performed subsequent to the vibration and shock tests. The Universal Standard Image Tube Tester Assembly F8584978 or Inspection Fixture F7560125 shall be utilized in this test for energizing the image tube and irradiating the photocathode of the tube. Energize the image tube with the voltage specified in 3.7.2. Irradiate the photocathode of the image tube. Adjust focus voltage to the image tube for best focus of the resolution target image at the center of the image screen, then remove the resolution target reticle from the field. A magnifying lens of at least ten power (10X) with a focal length that permits inspection of the lens and the image screen shall be utilized as an aid in visual inspections. A comparison standard that conforms to Drawing C7641866, Surface Quality Standards Comparison, shall be utilized for evaluating any surface defects of the field lens and image screen.

4.6.9.1 Field lens quality.- The field lens shall be visually inspected under 10X magnification for scratches digs or included bubbles. The surface quality of the lens within the 0.80 inch diameter circle shall not exceed the scratch and dig tolerance specified in 3.6.1. Bubbles shall be evaluated against the dig standard. The comparison standards specified in 4.6.9 shall be utilized for evaluating surface quality defects.

4.6.9.2 Screen lens quality.- The image screen (glass surface) shall be visually inspected under 10X magnification for scratch, dig, dirt and lint as specified in 4.6.9.1. The quality of the inner and outer zone of the image screen shall not exceed the applicable tolerance specified in 3.6.2.

4.6.9.3 Potting and pot cleanliness.- The pot shall be visually inspected under 10X magnification for evidence of loose dirt or lint, dust retention grease, potting resin defects, glass fracture or loose gaskets. The pot assembly shall conform to the requirements of 3.6.3. When it is necessary to ascertain that dirt or lint is movable, record position of the dirt or lint particle, then remove image tube from the test fixture and subject it to the low frequency vibration (4.6.7) for

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a minimum period of 30 seconds. Reinspect the recoreded positon for the particle of dirt or lint.

4.6.10 Conversion index. - The Universal Standard Image Tube Tester Assembly F8584978 shall be utilized for this test. The test procedure specified in 3.2 of the operating instructions on Drawing F8584978 shall apply, except that the face of the photocell cone frustrum shall fit snugly against the field lens aperture of the pot. The measured conversion index of the image tube shall not be less than the value specified in 3.8.1. Record the conversion index value for the following test. The image tube shall then be stored for a 672 hour holding period. Under the conditions set forth in 4.5.2.2 the Inspection Fixture F7560125 may be utilized for this test.

4.6.11 Conversion index decay.- The test equipment specified for measuring conversion index shall be utilized for measuring conversion index decay. Repeat the conversion index test at the completion of the 672 hour holding period, and record this second conversion index value, This second conversion index value shall not be less than 15. The conversion index decay of the image tube shall be computed as follows to determine acceptability of the tube:

a. Conversion index decay, 20 percent or less. - Compute the percentage difference between the first (4.6.10) and second (4.6.11) conversion index values. When the computed conversion index decay is 20 percent or less, the image tube shall be subjected to the tests for the remaining requirements of this specification.

b. Conversion index decay, maore than 20 percent. - When the computed index decay (4.6.11a) is more than 20 percent, the image tube shall be stored for an additional 336 hour period. Repeat the conversion index test in 4.6.10 and at the completion of this 336 hour holding period record the third conversion index value. The third conversion index value shall not be less than 15. Compute the percentage difference between the first (4.6.10) and third conversion index values. Where the computed conversion index decay is 30 percent or less, the image tube shall be subjected to the tests for the remaining requirements of this specifications A computed conversion index decay greater than 30 percent shall be cause for rejection for the image tube.

4.6.12 Conversion index to background.- The Universal Standard Image Tube Tester Assembly F8584978 shall be utilized for this test. Measure the background brightness of the image tube in accordance with the test procedure specified in 2.6 of the operating instructions on Drawing F8584978. Record the background brightness value for use in the ratio computation. specified in 3.8.3. The conversion index value used in this computation shall be the applicable value obtained in 4.6.lla or 4.6.11b. The conversion index to background shall not be less than the value specified in 3.8.3. Under the conditions set forth in 4.5.2.2 the inspection Fixture F7560125 may be utilized for this test.

4.6.13 Alinement. - Test Fixture Image Converter Tube D8258660 shall be utilized for this test. The test procedure for the image tube specified

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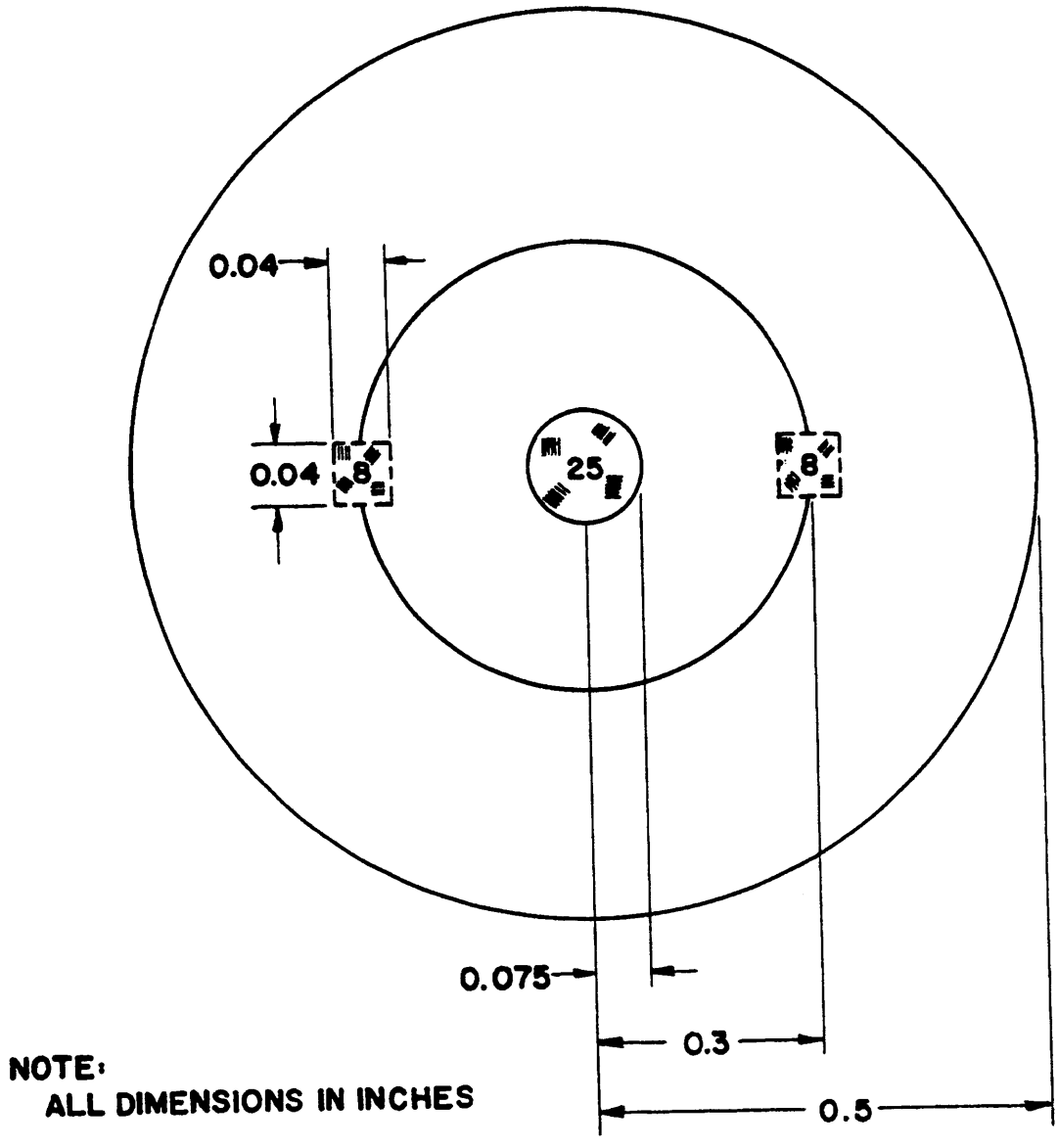
in 2.1 through 2.15 of the operating instructions on Drawing D8258660 shall apply, except that in place of applied decals, aperture stops shall be placed over the photocathode and field lens and two reference lines placed on the image tube. The reference lines on the image tube shall be two diameter lines positioned perpendicular to each other and shall be used as a rotational reference for any positioning of the image tube in the fixture. The field lens aperture stop shall be a disc having a geometrically centered 0.0375 inch maximm radius aperture placed into the ring to the field lens. The photocathode aperture stop shall be a disc having a geometrically centered 0.013 inch ± 0.001 inch diameter aperture placed into the metal ring of the photocathode end of the tube. The apertures of the discs shall be concentric with the gemetric axis of the tube. The alinement of the image tube shall conform to the requirements of 3.8.4, as determined by the position of the photocathode dot image with reference to the 0.075 inch aperture on the field lens. When less than 50 percent of the photocathode dot diameter is imaged within the 0.0375 inch radius aperture of the field lens the tube shall be unacceptable. Record position of dot image. The image tube that conforms to the requirement of 3.8.4 shall remain undisturbed in the fixture and immediately subjected to the test for the requirements of 3.8.5.

4.6.14 Image shift. - This test shall be performed in sequence to the alinement test utilizing Test Fixture D8258660. To test for image drift, energize the image tube for a minimum period of one minute while observing the photocathode dot image through the fixture microscope. During this one minute period the dot image shall show no evidence of image drift from its initial recorded position. Evidence of image drift shall be cause for rejection. The test for image shift shall be conducted in accordance with the test procedure specified in 2.16 through 2.20 of the operating instructions on Drawing F8258660, except that the vibration and shock tests shall be conducted in accordance with 4.6.7 and 4.6.8 of this specification. Image shift shall not exceed the tolerance specified in 3.8.5 during the application of electrical shock, before or after vibration and physical shock testing.

4.6.15 Resolution. - The Universal Standard Image Tube Tester Assembly F8584978 shall be utilized for this test. The test procedure specified in 2.9 of the operating instructions on Drawing F8584978 shall apply. The reticle pattern for the resolution test shall be constructed as shown in Figure I. Resolution of the image tube shall conform to the requirements of 3.8.6. Under the conditions set forth in 4.5.2.2 the Inspection Fixture F7560125 may be utilized for this test.

4.6.16 Cathode and screen spots.- The Universal Standard Image Tube Tester Assembly F8584978 shall be utilized for this test. The test procedure specified in 2.2 of the operating instructions on Drawing F8584978 shall apply. The quality of the photocathode and screen of the image tube shall conform to the requirements of 3.8.7. Under the conditions set forth in 4.5.2.2, the Inspection Fixture F7560125 may be utilized for this test.

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RESOLUTION PATTERN

FIGURE 1

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4.6.17 Magnification and focus.- Universal Standard Image Tube Tester Assembly F8584978 shall be utilized for this test. The magnification reticle, for center magnification shall contain two object points positioned at a radial distance of 0.0805 inch plus or minus 0.0001 inch from the geometric axis of the reticle. This reticle shall also contain two object points positioned at a radial distance of 0.3675 inch plus or minus 0.0001 inch from the geometric axis of the reticle for measuring percentage distortion. A resolution pattern of 25 line-pairs-per-millimeter shall be positioned at the geometric axis of the reticle to meet requirement 3.7.8. Position the geometric axis of the reticle on the geometric axis of the tube. The test procedure specified in 2.10 of the operating instructions on Drawing F8584978 shall apply. Center magnification of the image tube shall conform to the requirements of 3.8.8. Under the conditions set forth in 4.5.2.2 the inspection Fixture F7560125 may be utilized for this test.

4.6.18 Percentage distortion.- The test equipment specified in 4.6.17 shall be utilized for this test. The test procedure specified in 2.10 and 2.11 of the operating instructions on Drawing F8584978 shall apply. The percentage distortion of the image tube shall conform to the requirements of 3.8.9, as computed by the formula specified therein.

4.6.19 Cathode and screen uniformity.- The Universal Standard Image Tube Tester Assembly F8584978 shall be utilized for this test. Energize the image tube and adjust the focus voltage to the optimum value obtained in the resolution test. With no radiation incident on the photocathode, visually inspect the image screen for uniformity of background shading. Uniformity irradiate the photocathode within the lumen value specified, and visually inspect the image screen for uniformity of background shading. The background shading of the image screen, with and without radiation incident on the photocathode, shall conform to the requirements of 3.8.10. Under the conditions set forth in 4.5.2.2 the Inspection Fixture F7560125 may be utilized for this test.

4.6.20 Breakdown. - The Universal Standard Image Tube Tester Assembly F8584978 shall be utilized for this test. Perform this test with no radiation incident on the photocathode. Adjust the input voltage to the image tube to 17,000 volts, +0, -100 volts direct current (Vdc). Slowly sweep the focusing voltage from 2120 to 3375 Vdc while observing for any continuous arcing, spraying or flashing. The input focusing voltage shall not exceed 3375 Vdc during this test. The image tube shall conform to the requirements of 3.8.11. Under the conditions set forth in 4.5.2.2 the Inspection Fixture F7560125 may be utilized for this test.

4.6.21 Dark current. - The Universal Standard Image Tube Tester Assembly F8584978 shall be utilized for this test. The test procedure specified in 2.5 of the operating instructions on Drawing F8584978 shall apply. This test shall be conducted under the ambient humidity and temperature conditions specified in 3.8.12. The dark current of the image tube shall conform to the requirements of 3.8.12.

4.6.22 Irradiation surge. - The Universal Standard Image Tube Tester Assembly F8584978 and a standard time measuring instrument shall be

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utilized for this test. The tungsten light source shall be employed to produce, for no less than 0.5 second, irradiation of the image tube photo cathode sufficient to produce at least 3.5 uAdc. Over the specified period the image tube shall conform to requirement 3.8.13. Under the conditions set forth in 4.5.2.2 the Inspection Fixture F7560125 may be utilized for this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232 P7688738. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The image tube covered by this specification is intended for use in electro-optical infrared viewing devices which will provide a vision capability without the use of visible light.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification
b. Applicable packaging data sheet
c. Level of protection required

6.4 Definitions.

6.4.1 Filter factor (T).- The filter factor (T) is the fraction of the 2870°K hololumen flux transmitted by the infrared filter, and is defined as:

$$T = \frac{\int_0^\infty (R_\lambda J_\lambda T_\lambda) d\lambda}{\int_0^\infty (R_\lambda J_\lambda d\lambda}$$

where:

a. R_λ = The relative spectral response of an EIA-S1 photosurface
b. J_λ = The relative spectral distribution of the 2870°F color temperature light source
c. T_λ = The spectral transmission of the infrared filter
d. d_λ = The differential of λ
e. λ = The wavelength of the radiation

6.4.2 Filter. - This is one type of infrared filter glass designated as a Corning NO. 2540 (melt number 1613, 2.61 mm, in thickness) or equivalent.

6.4.3 Screen spots.

6.4.3.1 Ion spot. - A bright spot on the screen of the image tube caused by ionic bombardment when the photocathode is irradiated.

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6.4.3.2 Thermal or field emission spots .- Known as cold emission spots, streaks, and blemishes. They are of greater intensity than the background brightness of the image tube and are visible on the image screen with no radiation incident on the hotocathode. These spots on the image tube screen are the result of thermal or field emission.

6.4.3.3 Opaque or dark spots.- Imperfections visible on the image tube screen when the photocathode is irradiated.

6.4.4 Dark current.- Dark current is the electrical current which flows within the image tube and across the external surface of the tube with no radiation incident on the photocathode.

6.4.5 Type P-20 phosphor. - A substance used to coat the image tube screen which when bombarded by electrons will produce a yellowish-green visible light. It is of the low persistence variety. Information on this material will be found in the P-20 Phosphor Specification published by the Electronic Industries Association.

6.4.6 Geometric axis.- The geometric axis of the image tube shall be considered to be a line projected through the center of the 1.690 ± 0.005 inch diameter cathode ring and the center of the 1.990, ±0, -0.008 inch rabbeted diameter of the image tube pot assembly.

6.5 Date code instructions.- The date code specified in 3.8.2 shall consist of permanent legible lettering, a minimum of 1/8 inch high either on the glass or metal surface. The date code shall indicate the calendar month in abbreviated form, the last two digits of the calendar year, followed by the letters CI and value of conversion index obtained.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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MILITARY SPECIFICATION

ELECTRON TUBE: 7688738
(INFRARED, IMAGE CONVERTER)

This amendment forms a part of Military Specification MIL-E-60245(MU), dated 22 September 1965.

Paragraph 3.8.9 Percentage distortion: Delete "8 and 12 percent" and substitute "4 1/2 and 9 percent".

Paragraphs 4.2 and 4.2.1: Delete and substitute:

"4.2 First article (initial production] approval.- The requirement for first article approval and the responsibility (Government or contractor] for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 image tubes. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible fox conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, includig certificates of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

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Preparing activity:

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Project No. 1240-A258

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